Sequences and formal power series

The ordinary generating function (OGF) of $(a_n)_{n\geq 0}$ is $A(x) = \sum_{n\geq 0} a_n x^n$. With formal indeterminate x, neither numerical substitution nor convergence is required; coefficient comparison remains valid. Numerical real or complex evaluation additionally requires analytic convergence. Finite sequences give polynomials.

For a_n=1, $A(x) = \sum_{n\geq 0} x^n = (1-x)^{-1}$, meaning all coefficients agree. Formal addition and multiplication are defined algebraically. Division requires an invertible constant coefficient in the denominator, equivalent to a nonzero constant over a field.

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Which formal operations are legitimate?

A series is a coefficient sequence. Equality means equality at every degree. Thus $1/(1-x) = 1 + x + x^2 + \cdots$ first means $(1-x)(1 + x + x^2 + \cdots) = 1$, rather than a numerical limit. Convolution uses only finitely many terms at each fixed degree.

Assume coefficients belong to a field. For $A(x) = \sum a_n x^n$ with $a_0 \neq 0$, its inverse $B(x) = \sum b_n x^n$ is determined by

$$b_0 = 1/a_0, \qquad a_0 b_n + \cdots + a_n b_0 = 0, \qquad b_n = -\frac{1}{a_0} \sum_{k=1}^{n} a_k b_{n-k} \quad (n \geq 1).$$

Only earlier b coefficients occur, and the fixed nonzero denominator makes this recursion valid for every n.

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Which formal operations are legitimate?: derivation and application

An infinite product is legitimate when only finitely many factors affect each fixed coefficient. In $\prod_{k\geq 1}(1-x^k)^{-1}$, factors k>n equal one through degree n, satisfying this condition.

Formal power series

Compare coefficients

Add, multiply, differentiate formally

1/(1−x) is an inverse, not a limit

Operations follow formal algebra

Analytic functions



Check |x|<R before evaluation

Coefficient identities are formal; numerical values need analytic conditions

Numerically the geometric series converges for |x|<1. That condition is additional to, rather than a premise of, the formal coefficient identity.

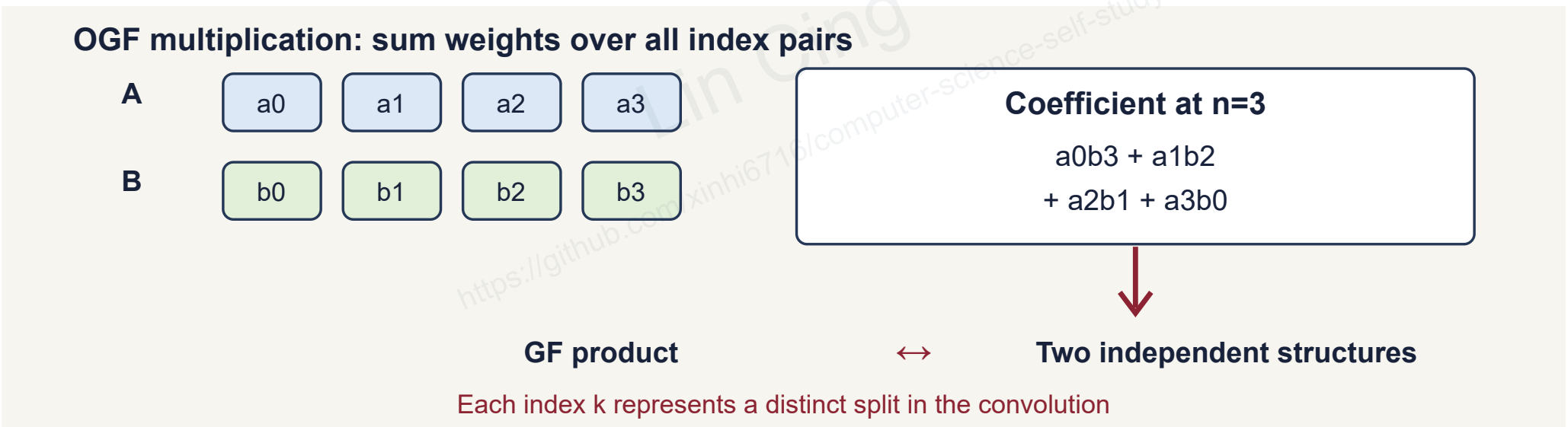
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Operations and convolution

For A and B with coefficients a_n,b_n,

$$A(x)B(x) = \sum_{n\geq 0}\left(\sum_{k=0}^{n} a_k b_{n-k}\right) x^n.$$

Multiplying by x^j shifts indices upward by j with j leading zeros; scalar multiplication scales coefficients. Convolution models a split into two independent smaller objects, summed over every possible split size.



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Operations and convolution: derivation and application

If objects have additive weight n and count a_n, define $A(x) = \sum a_n x^n$. Coefficients recover the counts once a counting rule or bijection is established. The product rule requires a unique decomposition into an ordered pair (u,v), with additive weights. Nonunique decompositions repeat objects. A partition factor $1/(1-x^k)$ records multiplicity of a fixed part size, so 3+1 and 1+3 do not differ; ordered compositions require a different model.

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Product models and coin combinations

For objects of unit weight, an allowed multiplicity set S gives factor $\sum_{r \in S} x^r$. If each object of this type has positive integer weight k, the factor is $\sum_{r \in S} x^{kr}$. Independent choices across types multiply; the degree-n coefficient counts total weight n. Coefficients are nonnegative integers counting permitted choices.

Unlimited coins of values 1,2,3 give $1/((1-x)(1-x^2)(1-x^3))$. Identical coins record quantities, not order. Ordered coin sequences instead use $D(x)=x+x^2+x^3$; length k gives $D(x)^k$ and all lengths, including empty, give $1/(1-x-x^2-x^3)$. At total three, unordered choices are 1+1+1,1+2,3; ordered sequences also include 2+1, giving four. The change is order, not invented labels on identical coins.

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Bounded choices and negative binomial series

Multiplicity at most m gives $1 + x + \cdots + x^m = (1 - x^{m+1})/(1 - x)$; exactly m gives x^m; a minimum a extracts x^a. Three types each allowed 0,1,2 have count $[x^n](1 + x + x^2)^3$. At n=5 only permutations of (2,2,1) contribute, giving three; unrestricted stars and bars $\binom{7}{2} = 21$ ignores the upper bound.

Four unlimited types give $(1 - x)^{-4} = \sum_{n \geq 0} \binom{n+3}{3} x^n$. Generally for positive integer r,

$$(1 - x)^{-r} = \sum_{n \geq 0} \binom{n + r - 1}{n} x^n.$$

The coefficient counts $y_1 + \ldots + y_r = n$ with nonnegative y_i. Arrange n stars and r−1 bars in $\binom{n+r-1}{r-1} = \binom{n+r-1}{n}$ ways, bijectively encoding all allocations. Requiring each type positive first places one of each, giving factor $x^r/(1 - x)^r$ and coefficient $\binom{n-1}{r-1}$ for $n \geq r$. Unlimited and bounded multiplicities require different factors and sample spaces.

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Bounded choices: remove the common baseline

At each of ten ordered positions choose one of $5, 8, 11, 14, 17$; count choices with total 80. Write the value at position i as $5 + 3t_i$, where $t_i \in \{0, 1, 2, 3, 4\}$. Positions are distinguishable, so component order matters.

At each position: start with 5, then add a multiple of 3

At position i choose $t_i \in \{0, 1, 2, 3, 4\}$
The corresponding value $5 + 3t_i \in \{5, 8, 11, 14, 17\}$

Baseline for ten positions: 10 × 5 = 50

Remainder: 80 − 50 = 30 = 3 × 10

Hence $t_1 + t_2 + \cdots + t_{10} = 10$; positions are distinguishable.

The total condition is $50 + 3\sum_{i=1}^{10} t_i = 80$, or $\sum t_i = 10$.

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Shift and extract the coefficient

Factor x^5 from each factor and put $y = x^3$:

$$(x^5 + x^8 + x^{11} + x^{14} + x^{17})^{10} = x^{50}(1 + x^3 + x^6 + x^9 + x^{12})^{10}.$$

Thus $[x^{80}]$ extracts the x^{30} coefficient of the parenthesized product, equivalently

$$[x^{80}](\cdots)^{10} = [y^{10}](1 + y + y^2 + y^3 + y^4)^{10}.$$

Choosing y^{t_i} from factor i corresponds bijectively to the choice at position i.

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Count the bounded coefficient by inclusion–exclusion

Without upper bounds, $t_1 + \cdots + t_{10} = 10$, $t_i \geq 0$, has $\binom{19}{9} = 92378$ solutions. Let A_i mean $t_i \geq 5$. For one specified violation, subtract 5; the remaining total is 5, giving $|A_i| = \binom{14}{9} = 2002$.

For two specified violations, subtracting 5 from each leaves total 0 and exactly one solution. Three violations require total at least 15, so their intersection is empty. Inclusion–exclusion gives

$$\binom{19}{9} - 10\binom{14}{9} + \binom{10}{2} = 92378 - 20020 + 45 = \boxed{72403}.$$

Each single-violation count includes double violations. The 45 restores those over-subtracted choices.

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Partial fractions and coefficient extraction

For example,

$$\frac{1}{(1-x)(1-2x)} = \frac{2}{1-2x} - \frac{1}{1-x} = \sum_{n\geq 0}(2^{n+1}-1)x^n,$$

with first coefficients 1,3,7,15. Preserve proper numerator degrees, all repeated-factor powers, and irreducible quadratic terms. A finite truncation cannot determine omitted coefficients.

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Partial fractions and coefficient extraction: derivation and application

For P(x)/Q(x), first divide polynomials if deg P≥deg Q. Factor Q over the reals into linear and irreducible quadratic factors. A repeated factor $(1-ax)^m$ requires every term $C_1/(1-ax)+\ldots+C_m/(1-ax)^m$. An irreducible quadratic normally requires a linear numerator. Extract using

$$[x^n](1-ax)^{-k} = a^n \binom{n+k-1}{k-1}.$$

An irreducible real quadratic may still simplify:

$$\frac{1}{1+x+x^2} = \frac{1-x}{1-x^3}, \qquad (c_0, c_1, c_2, \ldots) = (1, -1, 0, 1, -1, 0, \ldots).$$

The numerator 1−x is essential. Expanding $1-(x+x^2)+(x+x^2)^2-\ldots$ also works, but every term contributing to a fixed degree must be retained. Rational partial-fraction constants can combine into integer coefficients when the model requires integers.

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Formal differentiation and products

Differentiate $(1-x)^{-1} = \sum_{n\geq 0} x^n$:

$$(1-x)^{-2} = \sum_{n\geq 1} n x^{n-1} = \sum_{m\geq 0} (m+1) x^m.$$

Repeating r−1 times yields $(1-x)^{-r} = \sum_{m\geq 0} \binom{m+r-1}{m} x^m$. Formal differentiation only transforms coefficients and requires careful indices, not convergence. Analytic series and derivatives share a convergence radius but may differ at endpoints. Numerical comparisons require the convergence domain; finite polynomial truncations do not supply the missing coefficients. Low-degree expansion and stars-and-bars counts check the identity independently.

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Probability-generating coefficients

For nonnegative integer-valued X, define $P_X(z) = \sum_{k \geq 0} \Pr(X = k) z^k$. Coefficients are point probabilities, not densities. Nonnegative coefficients summing to one give P_X(1)=1.

For equal probabilities on 2,4,6,

$$P_X(z) = (z^2 + z^4 + z^6)/3, \qquad E[X] = P_X'(1) = (2 + 4 + 6)/3 = 4.$$

Also

$$P_X''(1) = (2 + 12 + 30)/3 = 44/3 = E[X(X - 1)],$$
$$E[X^2] = 44/3 + 4 = 56/3, \qquad \operatorname{Var}(X) = 56/3 - 16 = 8/3.$$

For infinite support, such as geometric distributions, differentiation and evaluation at one require justified limits or convergence and need not give a finite derivative. Formal coefficient identities alone do not establish finite moments; check support, normalization, and convergence.

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Integer partitions

The count p(n) of unordered positive-part sums has OGF

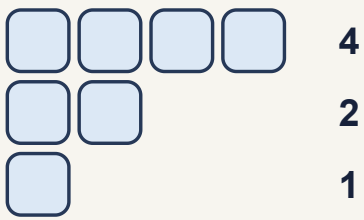
$$\prod_{k\geq 1}(1-x^k)^{-1}.$$

Each factor selects a multiplicity of part size k. Thus 3+1=1+3 as partitions. Ordered compositions, including empty, instead give $1/(1-(x+x^2+\cdots))=(1-x)/(1-2x)$; nonempty compositions give x/(1−2x).

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Integer partitions (continued)

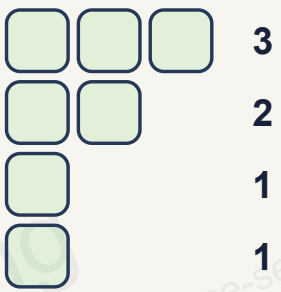
Ferrers transposition: a bijection with conjugate partitions



4
2
1

$7 = 4 + 2 + 1$

⇄



3
2
1
1

$7 = 3 + 2 + 1 + 1$

Same number of cells

Largest part 4 ⇄ 4 parts

3 parts ⇄ largest part 3

Equal family sizes

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At any degree n, only factors $k \leq n$ matter, making the infinite product formally valid. Fixed part-size factors remove order by retaining only multiplicity; their sequence in the product is fixed rather than permuting parts.

Bounds on the largest part

Largest part at most m gives $\prod_{k=1}^{m}(1-x^k)^{-1}$. At m=2, choose j twos and n−2j ones, for j=0,…,⌊n/2⌋, yielding ⌊n/2⌋+1 partitions.

Ferrers transposition exchanges largest part and number of parts. For $\lambda_1 \geq \cdots \geq \lambda_\ell > 0$, define $\lambda'_j = |\{i : \lambda_i \geq j\}|$. New rows are old column lengths, preserving total cells; $\lambda'_1 = \ell$ and the new number of parts is λ_1. A cell (i,j) exists iff $j \leq \lambda_i$ iff $i \leq \lambda'_j$; transposing again restores the original cells. This proves a reversible bijection, so at most m parts and largest part at most m are equinumerous. Distinct parts replace factors by $1+x^k$; at least one of a specified size k uses $x^k/(1-x^k)$. This can be imposed on finitely many specified sizes; no finite partition can contain every positive size.

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Odd parts and distinct parts

The product identity

$$\prod_{k\geq 1}(1+x^k)=\prod_{k\geq 1}\frac{1-x^{2k}}{1-x^k}=\prod_{k\geq 1}(1-x^{2k-1})^{-1}$$

comes from canceling even factors, proving equality of distinct-part and odd-part partition counts.

For an explicit bijection, write each odd part's multiplicity in binary. A nonzero bit 2^j creates one part of size 2^j times that odd part. Unique power-of-two-times-odd decomposition ensures distinct parts and a unique inverse.

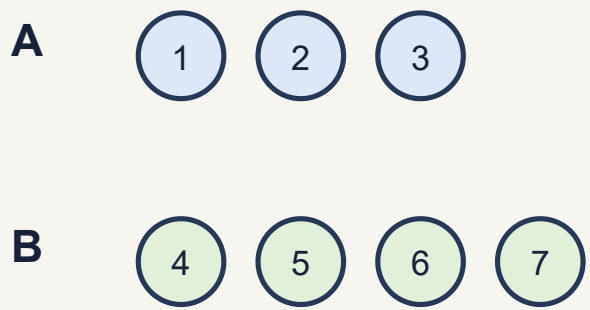
For 26=5+5+5+3+3+1+1+1+1+1, multiplicities 3=1+2,2=2,5=1+4 give {5,10},{6},{1,4}, hence 10+6+5+4+1. Reverse 10 to two fives, 6 to two threes, 5 to one five, 4 to four ones, 1 to one one. Odd families cannot collide because their odd factors are unique.

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Exponential generating functions

For labeled objects define $\widehat{A}(x) = \sum_{n \geq 0} a_n x^n / n!$. Splitting n=r+s labels into two groups gives $\binom{n}{r} = n!/(r!s!)$, which is exactly the normalization supplied by an EGF product.

EGF multiplication: split labels, then build each structure



A on r labels

times

B on s labels

Combined size n=r+s

Choose labels: C(n,r)

Label choices: C(n,r)=n!/(r!s!)

1/(r!s!) × n! restores the count

The normalized coefficient counts all label allocations

If a_r counts A-structures on any r labels and b_s B-structures on s disjoint labels, first choosing the A-label subset then both structures gives $\sum_{r=0}^{n} \binom{n}{r} a_r b_{n-r}$. Algebraically,

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Exponential generating functions (continued)

$$[x^n](\widehat{A}\widehat{B}) = \sum_{r=0}^{n} \frac{a_r b_{n-r}}{r!(n-r)!} = \frac{1}{n!} \sum_{r=0}^{n} \binom{n}{r} a_r b_{n-r}.$$

Multiplication by n! restores the count. The normalization encodes label splitting rather than an arbitrary conversion from an OGF.

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Exponential generating functions: derivation and application

Partitions into unlabeled nonempty blocks give $\exp(e^x - 1) = \sum B_n x^n / n!$. Initial Bell numbers are 1,1,2,5. Two labels either share one block or separate; three labels give one full block, three singleton-plus-pair choices, or three singleton blocks, totaling 1+3+1=5.

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Why SET components must be nonempty

Let $\hat{A}(0)=0$ count nonempty labeled components. Fixed k distinguishable slots have EGF $\hat{A}(x)^k$. Disjoint nonempty label sets make the components distinct objects, so every unordered component set has exactly k! slot orders. Therefore

$$\widehat{C}_k(x) = \widehat{A}(x)^k/k!, \qquad \widehat{C}(x) = \sum_{k\geq 0} \widehat{A}(x)^k/k! = \exp(\widehat{A}(x)).$$

At degree n only $k \leq n$ contributes. Empty components would allow arbitrarily many empty blocks and invalidate this interpretation and finite layering.

One nonempty set component has EGF e^x-1. For n=4,k=2, ordered sizes (1,3),(2,2),(3,1) contribute 4,6,4. Division by 2! gives seven, so $4![x^4](e^x-1)^2/2! = 7$. The four-label counts for k=1,2,3,4 are 1,7,6,1, totaling B_4=15.

Choose k companions of a new label in its block; partition all remaining labels. These reversible choices give

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Why SET components must be nonempty (continued)

$$B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k},$$

also obtained from $B'(x)=e^x B(x)$ by the EGF product rule.

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Repeated letters and parity restrictions

With positions as labels and letter i allowed multiplicities J_i,

$$\#\text{words} = n![x^n] \prod_i \left(\sum_{j \in J_i} x^j / j! \right).$$

For fixed multiplicities summing to n, the labeled product assigns positions in $n!/\prod j_i!$ ways, precisely the repeated-letter arrangement count. OGF multiplicity selection would ignore these orders.

With at most two A's, two B's, one C, length three gives

$$3! \, [x^3](1 + x + x^2/2)^2(1 + x) = 6(1 + 2) = 18.$$

Multiplicities (2,1,0),(1,2,0),(2,0,1),(0,2,1) each give three words; (1,1,1) gives six. Length four gives 4! (1/4+1)=30. Identical letters receive no artificial inventory labels.

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Repeated letters and parity restrictions: derivation and application

Even A-count, with B,C unrestricted, gives

$$n![x^n]\frac{e^x+e^{-x}}{2}e^{2x}=(3^n+1)/2.$$

The even filter cancels odd powers. At n=4, A-counts 0,2,4 contribute 16,24,1, totaling 41. Exponentials here are formal series, without numerical substitution.

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The summation operator

For $A(x) = \sum a_n x^n$ and $s_n = \sum_{k=0}^{n} a_k$, convolution with all ones gives

$$A(x)(1 + x + x^2 + \cdots) = \sum_{n \geq 0} \left(\sum_{k=0}^{n} a_k \right) x^n = S(x) = A(x)/(1 - x).$$

Thus 1/(1−x) is the sum operator; repeating it gives higher prefix sums. Weighted sums or a starting index of one require separate shifts. No analytic convergence is needed.

For a_k=2k+1, prefix sums are $(n+1)^2$, hence $A(x)/(1-x) = \sum_{n \geq 0} (n+1)^2 x^n$. Combining summation with x(d/dx) also derives power-sum identities.

For nonnegative u_1,u_2,u_3 with $u_1+2u_2+3u_3 \leq n$, add the unique nonnegative slack u_4 so $u_1+2u_2+3u_3+u_4=n$. The four factors give

$$\sum_{n \geq 0} A_n x^n = \frac{1}{(1-x)^2(1-x^2)(1-x^3)} = 1 + 2x + 4x^2 + 7x^3 + 11x^4 + 16x^5 + 23x^6 + 31x^7 + \cdots.$$

Thus A_7=31, agreeing with direct enumeration of the triples. The extra factor follows from slack, preserving the inequality's exact meaning.

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Fibonacci OGF and both closed-form derivations

For $F_0=0, F_1=1$ and $F_n=F_{n-1}+F_{n-2}$, sum shifted recurrences:

$$F(x) - x = xF(x) + x^2F(x), \qquad F(x) = x/(1 - x - x^2).$$

The denominator constant is one, so formal division is valid. The first coefficients are 0,1,1,2,3,5. Comparing $(1 - x - x^2)F(x) = x$ gives constant $F_0=0$, linear $F_1-F_0=1$, and for $n\geq 2$ exactly $F_n-F_{n-1}-F_{n-2}=0$, verifying every coefficient and initial shift.

Let $\varphi = (1 + \sqrt{5})/2, \psi = (1 - \sqrt{5})/2$, the roots of $r^2-r-1=0$. Then $1-x-x^2=(1-\phi x)(1-\psi x)$, $\phi-\psi=\sqrt{5}$, and

$$F(x) = \frac{1}{\sqrt{5}}\left(\frac{1}{1 - \varphi x} - \frac{1}{1 - \psi x}\right).$$

Its numerator is $((1 - \psi x) - (1 - \varphi x))/\sqrt{5} = x$. Extracting coefficients gives $F_n = (\varphi^n - \psi^n)/\sqrt{5}$.

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Fibonacci OGF and both closed-form derivations: derivation and application

Independently, the characteristic-root expression $F_n = A\varphi^n + B\psi^n$ has initial equations A+B=0 and Aϕ+Bψ=1, giving A=1/√5,B=−1/√5 and the same formula. Both routes need both initial values; roots alone do not specify the sequence. Formal identities hold for every n; analytic substitution requires |x|<1/ϕ. Chapter 10 applies the method to general recurrences.

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Boundaries and checks

Keep coefficient indices and polynomial degrees aligned, especially at n=0. Distinguish formal identities from analytic evaluation. Multiplicity limits, ordered sequences, and labels each change factors. Check extracted coefficients using a recurrence or small cases. The exercise and solution PDFs provide further derivations. A valid formula follows from a bijection or coefficient identity that can be expanded, rather than resemblance alone.

References and attribution

Lin Cing, handwritten discrete mathematics notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Ch. 9.

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